

Dhruv Rath

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SUMMARY

Data-focused B.Tech student experienced in transforming raw data into actionable insights using SQL, Python, data analysis and visualization, and predictive analytics. Skilled in developing end-to-end analytical solutions with additional experience in Machine Learning and Generative AI.

EDUCATION

Vellore Institute of Technology, Chennai

2023 – present (2027)

B.Tech - Electronics and Computer

CGPA - 9.06

Relevant Courses: Data Structures & Algorithm, OOP, Database Management, Operating Systems, Computer Networks, Computer Architecture, Cloud Computing, AI & ML, Deep Learning, Web Programming

EXPERIENCE

AI & ML Intern

June 2026 – August 2026

Sutherland

Chennai, Tamil Nadu

- Developed Python data pipelines and FastAPI services for LLM-powered retrieval and forecasting.
- Built data processing and analytical workflows supporting AI-driven predictions and decision-making.
- Architected a contractual-document RAG pipeline with hybrid retrieval, reranking, and citation-aware LLM generation; prototyped on 100 documents and designed to scale to 1M.

SKILLS

Languages: Python, SQL, Java, C/C++

Data Analytics: Pandas, NumPy, Data Cleaning, Data Transformation, EDA, Feature Engineering, Predictive Modeling, KPI Analysis

AI & ML: Scikit-learn, Regression, Classification, Feature Selection, Predictive Modeling, Deep Learning, RAG

Visualization & Tools: Matplotlib, Seaborn, Streamlit, MySQL, Excel, Git, GitHub

PROJECTS

SensorLens – Sensor Data Analytics Platform [GitHub](#) | [Live](#)

Python, Pandas, SQL, Streamlit

- Built an ETL pipeline for sensor data with validation, deduplication, rejection logging, quality reporting.
- Architected a normalized MySQL schema with advanced SQL analytics using CTEs, subqueries, window functions, and analytical views.
- Developed a Streamlit dashboard for KPI tracking, sensor trends, anomaly detection, and CSV export.

FinRisk – Financial Anomaly & Risk Narrator [GitHub](#)

Python, Pandas, NumPy, SQL, Streamlit

- Built a financial risk pipeline integrating data ingestion, transformation, validation, and multi-factor analysis.
- Implemented analytical workflows for anomaly detection, risk scoring, and extraction of market signals from structured financial data.
- Designed interactive dashboards for trend analysis, risk monitoring and financial insight generation.

PayCast – Invoice Payment Forecasting [GitHub](#) | [Live](#)

Python, Pandas, NumPy, Scikit-learn, Streamlit

- Designed an analytics system to forecast invoice clearing timelines across 50,000+ business transactions.
- Built an end-to-end pipeline for data cleaning, feature engineering, EDA, and RFE-based feature selection.
- Compared Random Forest, Gradient Boosting, and XGBoost, achieving R^2 0.656 and RMSE 8 days.
- Integrated the forecasting pipeline into a Streamlit dashboard for real-time predictions.

CERTIFICATIONS

- Infosys – Data Science
- Infosys – Machine Learning Implementation
- IBM – Advanced Generative AI - Professional

ACHIEVEMENTS

- Shortlisted among 416 of 3,01,066 teams (**Top 0.14%**) in the Adobe University Hackathon 2026.
- Design Lead for the Sports Club at VIT Chennai